
11.9 SOME USEFUL BOOKS

Chaubey, P. K. (2004), *Inequality: Issues and Indices*, Kanishka Publishers, Distributors, Delhi.

Cowell, Frank A. (1995), *Measuring Inequality*, Prentice Hall/Harvester Wheatsheaf, London;

Sen, A.K. (1997) *On Economic Inequality*, Oxford University Press, Oxford. (with Annexe by James E. Foster.

11.10 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Range is the difference between the maximum and minimum value. With a view to ensuring that the index of inequality based on this measure of dispersion is unit-free and/or is confined in the interval of (0,1), various ways of normalizations could be considered. See Sub-section 11.2.1.
- 2) In the relative range only extreme values are considered, which may not be representatives of the distribution. It is like comparing the poorest (who may be a few) with the richest (who may be one). Inter-quartile measures take into account the middlemost distribution with 50 percent recipients. See Sub-section 11.2.2
- 3) In the mean deviation, all values in a way are considered. The mean of absolute deviations is divided by the mean of the distribution to yield relative mean deviation. See Sub-section 11.2.3. Since the sum of deviations on one side of the mean will not change with the transfer of income contemplated, the magnitude of the index would not change.
- 4) See Sub-section 11.2.4. In one version the deviations are taken with respect to logarithm of geometric mean and in the other with respect to that of arithmetic mean. With some mathematical manipulation, one can find out that former is smaller than the latter by square of the difference between the logarithms of geometric mean and arithmetic mean.
- 5) It is a straight application of the fact that geometric mean of a distribution is smaller than its arithmetic mean. Of course, when the values are greater than 1!
- 6) Hirfindahl index is sum of squares of the shares with each recipient, which of course varies with the number of recipients. Equal distribution of shares between two recipients will yield a value of 0.5 and between three recipients, 0.333. It is therefore more used as a measure of concentration. See Sub-section 11.2.5.
- 7) The message is that one can try on one's own to devise new methods. See Sub-section 11.2.6. For second part, the answer is $5/9$. Try it out.

Check Your Progress 2

- 1) Gini ratio is one half of the average value of absolute differences between all pairs, including with self, of values divided by the arithmetic mean.

See the definition by Kendall and Stuart in Sub-section 11.3.1. Have a look at expression (g.4) and (G.7).

- 2) The basic difference lies Gini index from other measures based on dispersion that in articulating all differences are considered while in others either few differences are considered or deviations from arithmetic/geometric means are considered.
- 3) See Sub-section 11.3.2.
- 4) By writing out in a table, columns for class intervals or values, frequencies, class total values, cumulative frequencies, cumulative total values, cumulative proportions and cumulative shares in respect of each class. For using expression (LG.5), consecutive moving differences (or sums) of proportions and consecutive moving sums (or differences) need to be computed in two additional columns. MS-Excel will do well.

Check Your Progress 3

- 1) Look at the Fig. 11.4 and Sub-section 11.4.2. Try writing the properties in language.
- 2) When the two Lorenz curve will intersect, it will not be possible to say on balance which distribution is more unequal. In fact, one section in that case will be more unequal and the other section less unequal in distribution 1 in comparison to their counterparts in the distribution 2.
- 3) The value of Gini coefficient is equal to twice the areas inscribed between line of equality and the Lorenz curve. It is equivalent to saying that G is equal to $(1-A)$ where A is the areas below Lorenz curve in the unit box.
- 4) Kakwani's measure of inequality is normalized length of the Lorenz curve.

Check Your Progress 4

- 1) Dalton's social welfare function is a simple aggregation of welfare (utility) functions of the individuals constituting the society. In addition, all individuals are supposed to have the same income utility function.
- 2) Sub-section 11.5.1. Write Dalton's proposal and its modern version. Also point out that though conceived in terms of utility, Dalton held that the index has to be measured in terms of income only.
- 3) An inequality measure should not change with linear transformation of personal utility function. Since Dalton's index does not respect this property, Atkinson is not happy. He therefore devises a new artifact called 'equally distributed equivalent income' and suggests a new utility function called iso-elastic marginal utility function.
- 4) Sen index is different from the Atkinson's index in one respect that he opts for a social welfare function, which has as individual incomes its arguments and is symmetric and quasi-concave. See Sub-section 11.5.3.
- 5) See Sub-section 11.5.5.

AXIOMS OF INEQUALITY MEASURES

For any statistical measure, some of the desirable properties that are described in standard textbooks are (i) simplicity of comprehension, (ii) ease of computation, (iii) range of variation, and (iv) amount of information needed. However, we discuss below only those properties, which are peculiar to the measures of economic inequality.

We are often faced with situations where we have to compare two distributions with the help of an index with regard to their level of inequality. The two distributions may belong to two different countries at a point of time, to a country at two points of time, or to two situations—say, one before tax and the other after tax or before and after interpersonal transfers etc.

People have set up some intuitively appealing properties in order to judge the efficacy of an inequality index. The first set of properties was given as ‘principles’ by Dalton (1920). Today, in literature, they are known as axioms. We propose to discuss some common axioms. It may be pointed out at the outset that these axioms almost ignore the question whether inequality is an issue, which matters more (or less) in an affluent society or in a poor one. The whole discussion will assume that all incomes are positive though we know for sure in case of business failure or crop failure, incomes can well be negative or zero.

1) **Axiom of Scale Independence**

If there are two distributions of equal size such (N) that each element of one distribution is a multiple θ of the corresponding element of the other distribution, i.e.,

$$x_i^2 = \theta x_i^1 \quad i=1,2,\dots,N,$$

then the numerical magnitudes of inequalities of both the distributions should be the same, i.e.,

$$I(x_1^1, x_2^1, \dots, x_N^1) = I(x_1^2, x_2^2, \dots, x_N^2)$$

where the inequality measure I is shown as a function of the distribution

$$(x_1, x_2, \dots, x_N).$$

Obviously, it also satisfies the idea that the level of inequality should not change when the scale of measurement changes, say, from rupees to paise or bushels to quintals.

It does also mean that equal proportionate additions to all incomes would not change the level of inequality for

$$x_i(1 + \lambda) = \theta x_i \quad i=1,2,\dots,N,$$

The proportionate addition could even be negative. Thus, it is a question of shares in the cake, not the size of the cake. It is very obvious that an inequality

measure is defined in terms of shares s_i because a proportionate change in all incomes leaves the shares unchanged.

However, this axiom goes against Dalton's principle of proportionate additions to income, which stated that equal proportionate additions (subtractions) should diminish (increase) the level of inequality. Perhaps, Dalton could not see that, equal proportionate addition is theoretically, equivalent to change in the scale of measurement. It should so happen in the case of a measure of relative dispersion is obvious enough.

The axiom covers the cases of proportionate taxation/subsidies. It may be noted that such additions do not change individual (class) shares of total income and, therefore, the Lorenz curve remains unchanged. All measures based on the Lorenz curve shall therefore satisfy this axiom.

It should not mean that change in the size of cake is immaterial. In the social welfare, size of cake and distribution of cake both matter. It is only in a limited context of measurement of inequality that this property is considered desirable.

Lorenz (1997) mentions an objection raised against this axiom in terms of non-proportionate increase in well-being of different income holders, which means diffusion of well-being when incomes increase but concentration when incomes decrease. Thus, this idea existed much before Dalton mentioned it. One could easily see that this objection incorporates the idea of diminishing marginal utility. The true province of the axiom then is the unit of measurement.

2) Axiom of Population Size Independence

The level of inequality remains unaffected if a proportionate number of persons is added to each income level.

This suggests that the magnitude of inequality in the distribution of the cake should depend on the relative number of receivers with different levels of income. If we merge two economies of identical distributions of the same size N , then, in the consequent economy of size $2N$, there shall be the same proportion of the merged population for any given income. Such replications will leave the inequality level unchanged. The axiom is also known as the Principle of Population Replication.

This exactly corresponds to Dalton's principle of proportionate additions of persons. Since the Lorenz curve remains unchanged so long as proportions of people in each class remains the same, the measures based on the Lorenz curve would satisfy this axiom.

Let us have however a counter-intuitive example. Let us have a two-person world in which one person is having no income and the other having is having all. Let us replicate the economy. Now there is a four-person world in which two are sharing destitution with zero income but the other two are sharing positive income equally. Earlier there was no equality; now each 50 per cent of population is sharing the income equally. So, some scholars do not accept it.

3) Axiom of Equal Income Addition

If the distribution 2 $x_i^2, i=1,2,\dots,N$ is obtained by addition of equal amount d (say, through pension) to each element of distribution 1 $x_i^1, i=1,2,\dots,N$, i.e.,

$$x_i^2 = x_i^1 + d$$

then inequality level of distribution 2 should be lower than that of distribution 1 i.e.,

$$I(x_1^2, x_2^2, \dots, x_N^2) < I(x_1^1, x_2^1, \dots, x_N^1)$$

naturally, subtractions (say, taxation) of equal amount from each income would reverse the inequality sign. It can be noted that in the former situation, the shares of the poorer persons increase and in the latter, they decrease. This axiom exactly corresponds to Dalton's principle of equal additions to incomes.

Now, we propose to discuss two very important axioms relating to transfer of an income from a person to another when other things remain the same. The former may be called Pigou-Dalton condition and the latter, Sen condition.

4) First Axiom of Income Transfer (Pigou-Dalton Condition)

If an equal transfer from a richer person to a poorer person takes place, then the level of inequality is strictly diminished, provided that the equalizing transfer amount is not more than the difference between two incomes involved. Any number of such transfers taking place between any two consecutive income units will not cause any change in the ranking of income units and therefore such a process of transfers may be called the rank-preserving equalization.

This axiom requires an inequality measure to be sensitive to transfers at all levels of income and, thus, at least a function of all incomes.

This axiom corresponds to Dalton's principle of income transfer. Dalton (1920) argued that an inequality measure must have this minimal property. Since in this context Pigou's contribution (1912) is found significant, Sen (1973) designated this axiom as Pigou-Dalton condition. Following him, a number of contributors in the field have given it the name of 'P-D condition'.

Most of the indices, barring relative range and relative mean deviation, pass this test. This axiom is also known as weak transfer axiom because it suggests the direction but not the magnitude of change in the level of inequality.

5) Second Axiom of Income Transfer (Sen Condition)

If we consider two transfers, one at a time, at different points of scale, then the transfer at lower end of scale should have greater impact than its counterpart at higher end of the scale. According to Sen, (1973), the impact on the index should be greater if the transfer takes place from a person with an income level of, say Rs.1000 to someone with Rs.900 than a similar transfer from a man with Rs.1000100 to someone with Rs.1000000.

We may see many measures do not satisfy any of the two conditions of transfer and some satisfy only the first one. Those that satisfy the second transfer axiom automatically satisfy the first transfer axiom.

6) Axiom of Symmetry

If distribution $(x_1^p, x_2^p, \dots, x_N^p)$ were a permutation of distribution $(x_1, x_2, \dots, x_N)$, then the inequality level of both the distributions would be the same.

This implies that if two persons interchange their income positions, inequality measure does not change. Thus the axiom ensures impartiality between individuals for non-income characteristics: The evaluator does not distinguish between Amar, Akabar and Anthony; nor between Shiela and Peter; or between Mr. Pygmy and Ms. Dwarf. Further, it means that the inequality depends only on the frequency distribution of incomes.

7) Axiom of Interval

The inequality measure should lie in the closed interval of $(0,1)$.

The measure is supposed to assume the value of zero when all incomes are equal, which means when all persons have equal income and the value of unity when only one individual gets all the income (and others have zero incomes, not negative incomes).

Most people tend to agree with the axiom. A few, notably Theil (1967) and Cowell (1995), disagree. They hold that the situation of one person grabbing all the income in a society of 2 persons cannot be described by the same level of inequality as that of one person doing so in a society of 2 crore persons. It would not be easy to assert that in the case of 2-person society the level of inequality is unity when one person has all the income and the other has none. In one case, 50 percent population is having positive income, in the other only 0.00000005 percent. Some people therefore qualify the axiom by saying that when one person gets all the income the measure approaches unity in the limit as the number increases.

When a measure has a finite maximum, it is easy to transform such an index into the one, which has maximum value 1. Most measures, though not all, have zero as their minimum value. But question that Cowell (1995) raises is that there are many ways in which the measure could be transformed so that it lies in the zero-to-one range but each transformation has different cardinal properties.

8) Axiom of Decomposability

Suppose population can be sub-divided into several groups and an over-all index of inequality was a function of group-wise indexes and if the population mean can be expressed as weighted average of group means, the population index of inequality can be regarded as decomposable. The groups might be defined as comprising of people in different occupations, residents of different areas, with different religious or educational backgrounds etc.

However we find a lot of overlapping in these groups. This leads sum of the weights to differ from unity.

Not all indices are found to be decomposable. Gini coefficient, a very popular measure is decomposable only if the constituent groups are non-overlapping. Cowell (1995) has conducted a beautiful experiment. First, he computes four inequality measures for two distributions of same size and same mean—each divided into two groups of equal size in a manner that there is no overlapping:

Population A: (60,70,80), (30,30,130)

Population B: (60,60,90), (10,60,120)

Now, it is found that the group means and population means in two distributions are the same and group inequalities in B are higher than their counterparts in A. But when we compute overall inequalities, one of the measures suggests that the magnitude of inequality in B is lower than that in A. And the measure used is Gini, which is very popular among economists. As he says, 'strange but true'. If the component inequality magnitudes are higher and the weights are the same, how could overall measure be lower? It is therefore impossible to express overall inequality (change) as some consistent function of inequality change in the consisted groups.

These are all intuitively appealing axioms. There does remain scope for formulating other axioms. In the literature on poverty measurement one finds a plethora of axioms developed by a number of contributors working in that areas. But we shall be content with these only.



